Lab 2 Rubric: Free Fall from a Ticker Tape

Out of 100 points. This is the rubric on the LB2-M1 assignment in Brightspace. Lab 2 page · Word copy · PDF copy

Use this rubric as your writing guide: the criteria follow the lab report template top to bottom. Full-credit descriptions tell you what a finished section contains.

Grammar, spelling, and formatting problems are assessed 1 point per occurrence. Deadline: a report submitted by the deadline may be revised and resubmitted up to two more times. A report that arrives after the deadline but before grades are posted is graded as on time but cannot be revised; one that arrives after grades are posted is late, may take a penalty, and gets no detailed feedback.

The lab is scored out of 100 on the order-2 fit alone. The optional velocity method in the last section of the handout is worth up to 3 bonus points on top of that; skipping it costs you nothing.

Criterion	Pts	Complete	Strong	Developing	Needs work	Missing
Heading	3	**3** Names, course and section, lab number, and both dates present, properly formatted.	**2** One item missing or misformatted.	**2** Template placeholders or brackets left in the heading, or the section number missing.	**1** Several items missing.	**0** No heading block.
Abstract	7	**7** Four sentences or fewer; states the lab's theme, the measured g with its uncertainty, and the % error. Nothing more.	**6** All content present but runs long, or one required number missing.	**4** Reports per-tape values but never a single measured g with an uncertainty, or the % error is omitted.	**2** Describes the lab but reports no numerical result.	**0** No abstract.
Introduction	10	**10** States $x = x_0 + v_0\Delta t + \frac{1}{2}a(\Delta t)^2$ and the generic parabola $y = Ax^2 + Bx + C$, defines every variable with units, explains why a fixed-frequency timer is used instead of a stopwatch, and ends with the question the lab answers.	**8** Equations present but a variable undefined, or the guiding question missing.	**6** One of the two equations is missing, or a variable is defined incorrectly, though the guiding question is present.	**3** Background only, with no equations, or no guiding question.	**0** No introduction.
Experimental: Materials & Setup	8	**8** Timer frequency stated, read off the unit used; hanging mass and drop height given; setup photo and a labelled sketch showing the position of dot 0 and the positive direction.	**7** Timer frequency, hanging mass and drop height all given, but the sketch or photo is unlabelled or unclear.	**5** Apparatus and timer frequency described in prose, but the setup photo and labelled sketch are missing, or the hanging mass and drop height are not given.	**3** Bare equipment list, or the timer frequency is not stated.	**0** No materials or setup description.
Experimental: Procedure	8	**8** Written in own words as prose; describes threading and releasing the tape, how dot 0 was chosen, that distances were measured from dot 0 rather than dot-to-dot, and that three tapes were run; reproducible blindly.	**7** Reproducible and in the student's own words, but one detail is thin.	**5** Reproducible, but missing the dot-0 criterion or the measure-from-origin detail.	**3** Copied or closely paraphrased from the handout.	**0** No procedure.

Criterion	Pts	Complete	Strong	Developing	Needs work	Missing
Results: Data Tables	12	**12** Raw dot data for all three tapes (dot number, time, distance in cm, position in m) plus a fit-results table carrying A, B, C, R^2, g = 2A and the % error. Every table numbered, captioned, units on every column, and introduced with a "Table X shows…" sentence.	**10** All tables present; minor issues — a caption or lead-in sentence missing, one unit omitted.	**7** Fit-results table complete, but the raw dot data is missing or covers fewer than three tapes.	**4** Units widely absent, or only partial data presented.	**0** No data tables.
Results: Graphs	9	**9** One position-vs-time graph per tape, three in all, each with an order-2 trendline whose equation and R^2 are displayed on the chart, numbered as a Figure, captioned, and described in the text.	**7** All three present; equation or R^2 missing from one, or a caption missing.	**5** All three present but trendlines shown without their equations, or the figures numbered as Tables.	**3** Fewer than three tapes graphed.	**0** No graphs.
Results: Sample Calculations	13	**13** One worked sample of each type, with units carried at every step: cm→m conversion, dot number to time (t = n/f), a = 2A, the mean, the standard deviation, and the % error.	**11** All types present with minor errors in math, rounding, or units.	**8** Four or five of the six types worked; one or two missing.	**4** Only one or two types shown, or units generally absent.	**0** No sample calculations.
Discussion	15	**15** Opens with the measured value ± uncertainty; states the direction of the miss and names a physical mechanism consistent with that direction, quantified if the error exceeds 10%; interprets B physically as the speed at dot 0 and converts it to a time into the fall; addresses what R^2 does and does not establish; acknowledges that N = 3 limits the uncertainty claim. Never says "human error."	**12** Most of the above; one element thin — usually the quantified error source or the interpretation of B.	**9** Opens with a result and interprets it, but two or three of the required moves are absent — the direction of the miss and its mechanism, the interpretation of B, what R^2 does and does not establish, or the N = 3 limitation.	**5** Restates the results without interpreting them, or attributes error vaguely.	**0** No discussion.
Additional Questions	5	**5** All three Additional Questions (AQ1–AQ3) answered in paragraphs, in a dedicated section.	**4** One of the three answered thinly.	**3** Answered inside another section rather than under their own heading.	**2** Answered as a bare numbered list, or several missing.	**0** Not answered.
Conclusion	5	**5** Three to four sentences; restates g with its uncertainty and % error, gives a definitive statement of the finding, and names one concrete improvement with a sense of how much it would help.	**4** Present but missing the numbers or the improvement.	**3** Present with the numbers but no definitive verdict, or no improvement named.	**2** Present but vague or discussion-like.	**0** No conclusion.
References	5	**5** At least one source beyond the textbook and this handout, on g and how it varies with latitude and altitude; author, title and date given for each; the handout itself cited.	**4** Multiple references but formatting errors, or a bare URL.	**3** Textbook and handout only, properly formatted.	**2** Handout only, or textbook only.	**0** No references.
Total	100					
Optional — velocity method (bonus)	+3	**+3** Velocity-vs-time graph for at least one tape, numbered and captioned, with its linear fit equation and R^2; one central-difference velocity as a sample calculation with units; two or three sentences comparing it to the order-2 result.	**+2** Graph and comparison present, sample calculation missing or unitless.			

Overall: Complete 90+ · Nearly there 75+ · Needs work 50+ · Missing below 50.

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